ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Teaching and answer guidance · Paver operation

Volume 6 · I1, I2, I3, I4, I5, I6, I7, I8, I9

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Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for paver operation. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS
Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT
Audio file / video container / timed web captions; test playback and retain a transcript.

Asphalt placement process

Worked example · Convert a trial thickness relationship

A fictional trial produced 50 mm compacted thickness from 62 mm loose thickness. The same approved trial relationship is used to estimate a 45 mm compacted target.

Loose/compacted factor = 62/50 = 1.24. Estimated loose thickness = 45 × 1.24 = 55.8 mm. This is an exercise estimate tied to the stated trial, not a universal setting; confirm with actual placement and compaction checks.

Explain and apply the idea

Put surface acceptance, treatment readiness, paver setup, placement and rolling in a useful sequence.

Answer guidance.

Accept and prepare the surface; apply/verify the specified treatment; inspect and set the paver/screed; place consistently; compact through the approved rolling plan. Communication and quality checks continue throughout.

Observe: Reading, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

A rectangular 50 m × 3 m mat has 0.05 m compacted thickness

A rectangular 50 m × 3 m mat has 0.05 m compacted thickness. Find compacted volume.

Answer guidance.

50 × 3 × 0.05 = 7.5 m³. Tonnage requires an appropriate density; loose thickness and delivered mass are different quantities.

Observe: Numeracy through the learner's actual contribution, calculation, explanation or saved work.

Write a call to the roller lead when delivery is interrupted.

Write a call to the roller lead when delivery is interrupted.

Answer guidance.

State paver location, expected interruption, placed area needing attention, known temperature/quality information and proposed coordination. Confirm the rolling plan with the lead rather than inventing new pass counts.

Observe: Communication, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Explain the complete placement sequence and treatment distinction.	0 / 1 / 2	Evidence/help: _____
Verify starting support, line and trial-based thickness allowance.	0 / 1 / 2	Evidence/help: _____
Coordinate delivery, placement and compaction with clear role boundaries.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Paving machine equipment and attachments

Worked example · Two sensor jobs

A diagram labels one sensor near the auger and another following a reference beside the machine.

The feed sensor can control the material supply in its assigned zone; a grade sensor follows a reference for elevation control. Confirm the actual model and function from the manual. Similar-looking sensors do not necessarily control the same system.

Label ten parts on the supplied paver figure and state each function.

Label ten parts on the supplied paver figure and state each function.

Answer guidance.

Use hopper, wings, push rollers, conveyors, augers, tracks/wheels, operator station, tow arms, screed plate and end gates or other correctly identified visible parts. Explain functions using the diagram and model scope.

Observe: Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

Convert nominal 8 ft and 10 ft widths to metres

Convert nominal 8 ft and 10 ft widths to metres. Do these establish the actual screed settings?

Answer guidance.

8 × 0.3048 = 2.4384 m; 10 × 0.3048 = 3.048 m. These conversions do not establish a particular machine’s working width or setting.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Ask an AI to create a component table from the supplied figure and caption

Ask an AI to create a component table from the supplied figure and caption. How will you audit it?

Answer guidance.

Require visible labels, functions and source/page references; verify every row against the image/manual. Mark hidden or unclear parts as unconfirmed. Do not accept invented labels or model-specific settings. Save the checked table with its source credit.

Observe: Digital, Writing, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify tractor and screed assemblies accurately.	0 / 1 / 2	Evidence/help: _____
Distinguish feed, grade and slope-control functions.	0 / 1 / 2	Evidence/help: _____
Check attachment compatibility and document scope.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Paver work safety procedures

Worked example · A stopped auger

Material stops moving, and a worker reaches toward the auger to remove a lump.

Stop the approach and keep clear. A pause in automatic feed does not isolate the auger; it may restart. Use the approved stopping, lockout, stored-energy and clearing procedure with authorized personnel.

Name the safe response when the truck guide’s signal is lost.

Name the safe response when the truck guide’s signal is lost.

Answer guidance.

Stop the affected movement, maintain safe positions and re-establish acknowledged communication before resuming.

Observe: Communication, Collaboration through the learner’s actual contribution, calculation, explanation or saved work.

A revised layout moves the paver edge 0.4 m toward a public route

A revised layout moves the paver edge 0.4 m toward a public route. What must be checked before proceeding?

Answer guidance.

Reassess the actual clearance, barriers, swept paths and access plan with the responsible lead/TCP team. The numerical shift alone does not establish acceptable separation.

Observe: Numeracy, Problem solving, Adaptability through the learner’s actual contribution, calculation, explanation or saved work.

Prepare a three-field defect note for a damaged hopper guard.

Prepare a three-field defect note for a damaged hopper guard.

Answer guidance.

Identify machine/location and damage; state held/out-of-service status and immediate controls; name the person notified and required authorized correction/release.

Observe: Writing, Digital through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use selected PPE and identify machine-specific hazards.	0 / 1 / 2	Evidence/help: _____
Demonstrate full approved lockout and verification.	0 / 1 / 2	Evidence/help: _____
Secure the machine with required supports, parking controls and access protection.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Asphalt paver operator roles and responsibilities

Worked example · Hopper level is not the whole story

The hopper contains mix, but the right side of the screed has a visibly reduced material head and texture changes.

Report the side and location, coordinate with the screed operator and inspect the feed-system status from safe positions. A full hopper does not prove uniform delivery across the screed. Use authorized adjustments and record the response; do not reach into the feed system.

Explain the difference between hopper contents and head of material.

Explain the difference between hopper contents and head of material.

Answer guidance.

The hopper receives/stores incoming mix; conveyors and augers deliver/distribute it to the screed. Head refers to the material in front of the screed. One does not guarantee the other is consistent.

Observe: Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

A 20 t load lasts 8 min at a steady fictional rate

A 20 t load lasts 8 min at a steady fictional rate. Find t/min; why is it not a universal paver speed?

Answer guidance.

20/8 = 2.5 t/min. Travel speed also depends on width, thickness, density, supply and the approved process; mass rate alone is insufficient.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Draft a truck-change handover that includes an access constraint.

Draft a truck-change handover that includes an access constraint.

Answer guidance.

Identify machine/lane, designated guide, approach/stop point, protected access, overhead concern, communication channel and lost-signal response. Confirm read-back before movement.

Observe: Writing, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Align and follow a verified reference.	0 / 1 / 2	Evidence/help: _____
Coordinate truck transitions and consistent material head.	0 / 1 / 2	Evidence/help: _____
Recognize and respond to obstacles and communication loss.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Paving machine maintenance

Worked example · A repeated low level

Hydraulic oil is low again after yesterday's top-up. A fresh wet area is visible beneath the machine.

Record the repeated loss and safe visual observation, secure/hold the machine as required and notify the responsible person. Do not treat repeated top-up as a repair or touch the suspected leak. Confirm authorized correction before release.

A fictional service interval is 10 operating hours

A fictional service interval is 10 operating hours. Last service 1,420 h; current 1,429.5 h. How much remains?

Answer guidance.

9.5 h elapsed; 0.5 h remains under this exercise interval. Calendar, condition and other schedule requirements may make service due sooner.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

Explain why an unlabelled spray bottle cannot be accepted as a release agent.

Explain why an unlabelled spray bottle cannot be accepted as a release agent.

Answer guidance.

Its identity, compatibility, hazards and approved use cannot be confirmed. Obtain the labelled approved product and SDS before use.

Observe: Problem solving, Communication through the learner's actual contribution, calculation, explanation or saved work.

Create a digital maintenance record without claiming checks you did not perform.

Create a digital maintenance record without claiming checks you did not perform.

Answer guidance.

Include machine/serial, meter, date, actual checks/service, lubricant/fluid identity, defects, status and responsible person. Mark unobserved or not applicable with reason.

Observe: Digital, Writing, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Service lubrication/intake/separator according to the actual schedule.	0 / 1 / 2	Evidence/help: _____
Check correct fluids and accessible cooling/battery components safely.	0 / 1 / 2	Evidence/help: _____
Clean feed and access areas with containment and restore guards.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Markers, grades and stakes

Worked example · Offset versus edge

A stake marked 1.5 m RT OFFSET describes the approved right paving edge at station 0+060. A learner treats the stake itself as the edge.

Use the project legend and stated offset direction to locate the actual edge; the stake is a reference away from it. Verify the relation with the grade lead before setting the paver line. Do not shift the whole mat to the stake.

At one verified point, base EL is 101.250 m and finished EL is 101.300 m

At one verified point, base EL is 101.250 m and finished EL is 101.300 m. Find the vertical difference.

Answer guidance.

0.050 m = 50 mm. It is the specified geometric difference at that point, not automatically the loose screed setting.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Write a read-back containing station, side, offset and unresolved datum.

Write a read-back containing station, side, offset and unresolved datum.

Answer guidance.

Example: “Station 0+060, right-side offset stake, marked 1.5 m. I need confirmation of the reference direction and datum before using its elevation.” Include the actual mark and ask the responsible person to confirm.

Observe: Writing, Communication through the learner’s actual contribution, calculation, explanation or saved work.

Communicate and verify the work

Use an AI to explain the abbreviation list, then identify what it cannot establish from the list alone.

Answer guidance.

It can draft definitions, which must be checked against the project legend. It cannot establish actual utility depth, property boundary, datum relationship or safe machine clearance from abbreviations alone.

Observe: Digital, Reading, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Interpret and protect the actual control marks.	0 / 1 / 2	Evidence/help: _____
Transfer line/elevation information correctly.	0 / 1 / 2	Evidence/help: _____
Maintain smooth alignment while coordinating truck and screed roles.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Operate asphalt paver

Worked example · Flat practice is a specific achievement

A learner completes a controlled flat-pad pass with simulated material. The next job involves a steep access road and hot mix.

Record the demonstrated flat-pad skills. Plan the additional machine/site-specific preparation and supervised workplace assessment for grades, slopes and hot-mix conditions. Do not label the first exercise as proof of unrestricted operation.

List five observations an instructor should record during a flat-pad pass.

List five observations an instructor should record during a flat-pad pass.

Answer guidance.

Examples: prechecks; safe entry/exit; surroundings checks; smooth forward/reverse stops; correct parking controls; hopper/feed control; alignment; acknowledged truck/screed communication. Record actual behaviour and help.

Observe: Writing, Reading through the learner’s actual contribution, calculation, explanation or saved work.

A fictional 36 m practice pass takes 12 min

A fictional 36 m practice pass takes 12 min. Calculate average travel speed and one reason not to compare learners by it alone.

Answer guidance.

3 m/min. Different support, material, interruptions and access needs affect time; safe control and specified quality are the criteria.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

During the pass, a partner reports a person entering the work area

During the pass, a partner reports a person entering the work area. Demonstrate the response.

Answer guidance.

Stop affected movement safely, communicate the intrusion and maintain control until the area and plan are restored. Resume only after clear acknowledgement.

Observe: Communication, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Precheck, maintain, start and monitor the actual machine.	0 / 1 / 2	Evidence/help: _____
Enter/exit safely and perform flat forward/stop/reverse/stop with correct parking controls.	0 / 1 / 2	Evidence/help: _____
Operate hopper, transmission and flow controls under supervision.	0 / 1 / 2	Evidence/help: _____
Position and guide the truck, follow the line and coordinate the screed/ground crew.	0 / 1 / 2	Evidence/help: _____
Place an approved simulated or asphalt mat, then clean and shut down.	0 / 1 / 2	Evidence/help: _____
Record additional workplace grades/slopes/conditions separately.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Follow shut down procedures

Worked example · An incomplete handover

The paver is parked correctly, but the operator noticed a leak and left without recording it.

The shutdown/handover is incomplete. Record the leak observation, location and machine status, notify the responsible person and arrange the required hold/inspection. The next operator needs the information before starting.

Explain shutdown versus lockout.

Explain shutdown versus lockout.

Answer guidance.

Shutdown ends operation in the prescribed state. Lockout isolates, secures and verifies hazardous energy for work requiring protection. A shutdown may be part of lockout but is not the whole process.

Observe: Reading, Communication through the learner's actual contribution, calculation, explanation or saved work.

A ten-item assessment gives 17/20 but a required screed-support check failed

A ten-item assessment gives 17/20 but a required screed-support check failed. State the result.

Answer guidance.

85% numerically, but not released: the safety-critical failure must be corrected and reassessed.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Prepare a concise next-shift note.

Prepare a concise next-shift note.

Answer guidance.

Include machine/meter, location and parking/support state, completed cleaning/service, defects/hold status, keys/access arrangement and person responsible for the next action.

Observe: Writing, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Choose and secure the approved parking location.	0 / 1 / 2	Evidence/help: _____
Complete model-specific shutdown and component support.	0 / 1 / 2	Evidence/help: _____
Inspect safely and hand over defects and status.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Transport paver

Worked example · Width changes with configuration

A fictional machine measures 2.50 m with transport equipment retracted and 3.10 m with a fitted extension assembly.

The actual configured width differs by 0.60 m. Verify the required transport configuration and current route/permit conditions with the carrier. The smaller dimension cannot be used on paperwork while the larger configuration remains fitted.

Name four checks before the machine touches the ramp.

Name four checks before the machine touches the ramp.

Answer guidance.

Correct OMM/configuration and mass; carrier/ramp rating and support; alignment/traction/clear area; designated operator/guide with acknowledged signals. Include route/permit preparation within the transport plan.

Observe: Reading, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

A base machine is 8,600 kg and fitted items add 740 kg

A base machine is 8,600 kg and fitted items add 740 kg. Find the load before separate cargo.

Answer guidance.

9,340 kg. Verify actual configuration and all carrier, axle, ramp and securement limits; this arithmetic is not a transport approval.

Observe: Numeracy through the learner’s actual contribution, calculation, explanation or saved work.

Write a handover when loading is delayed by an unidentified tie-down point.

Write a handover when loading is delayed by an unidentified tie-down point.

Answer guidance.

Hold loading/securement as appropriate, identify the machine and disputed point, obtain the matching manufacturer information and carrier confirmation, then revise and brief the plan before continuing.

Observe: Writing, Communication, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Verify dimensions, transport rules and assigned roles.	0 / 1 / 2	Evidence/help: _____
Prepare screed/attachments to the actual OMM.	0 / 1 / 2	Evidence/help: _____
Demonstrate controlled loading preparation and rated securement checks.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Integrated practice · all nine Skills for Success

Keep delivery and mat records consistent

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional approved area is 50 m × 3.2 m with a 45 mm compacted target. A verified trial relationship is loose/compacted = 1.24 for this exercise. The delivery register contains ticket 81 for 12 t and ticket 82 for 11 t, but a draft totals 35 t by counting ticket 81 twice. No density for estimating required mass is supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Evidence and feedback: Use the exact case facts; distinguish observation, assumption and missing authority.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Evidence and feedback: The note preserves verified facts, units and unresolved holds without inventing a release.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Evidence and feedback: Use the worked numerical key below; show the relationship and its practical limits.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Evidence and feedback: Record the product/capabilities, prompt, source facts, saved files and checks. A chat response alone is not a created or verified file. Check formulas and preserve originals.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Evidence and feedback: Separate the observed issue from a guessed cause; preserve evidence and identify the correct decision role.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Evidence and feedback: Observe an actual exchange and confirmation of the essential facts; do not infer communication from a shared file.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Evidence and feedback: Both learners take an identifiable turn; feedback is respectful and tied to the case evidence.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Evidence and feedback: Use the change-response key below; retain the earlier version and do not alter unrelated facts.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Evidence and feedback: The visual/media has accurate labels, source credits and a stated purpose. Inspect generated diagrams/media; do not invent technical settings. Keep a transcript/captions when producing audio/video.

Change card · Use after the first attempt

The coordinator asks for an audio handover as well as an editable table and PDF.

Verified case key

Area 160 m²; compacted geometric volume 7.2 m³; trial-based loose estimate 55.8 mm. Unique delivered tickets total 23 t, not 35 t. Do not infer required mass or acceptance without the missing density/criteria.

Change-response key

Create a checked script from the verified facts, identify the actual narration tool, retain transcript and inspect the audio; keep all file versions consistent.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person’s actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.